Multiple variables - mixed practice

Mix two variables, three variables, squares and products.

Name: _____ Suggested time: 10 minutes

Show your working. Use brackets when substituting negative values.

01 Given x=6, y=5

Evaluate: 6x+5y

02 Given a=-2, b=-1, c=0

Evaluate: 4a - 4b +3c

03 Given a=2, b=-3

Evaluate: a^2 - 6ab + b^2

04 Given x=2, y=1

Evaluate: 5x+2y

05 Given a=-3, b=-4, c=1

Evaluate: 2a - 6b +2c

06 Given a=-3, b=1

Evaluate: a^2 - 6ab + b^2

Answer sheet

Multiple variables - mixed practice | 6 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 61

02 -4

03 49

04 12

05 20

06 28

Worked solutions

Multiple variables - mixed practice

01 Given x=6, y=5

$6x+5y$

Substitute x=6, y=5.

$6\times 6 +5\times 5$

$36 +25$

61

02 Given a=-2, b=-1, c=0

$4a - 4b +3c$

Substitute values.

$4\times (-2) - 4\times (-1) +3\times 0$

$-8 +4$

-4

03 Given a=2, b=-3

$a^2 - 6ab + b^2$

Substitute a=2, b=-3.

$(2)^2 - 6(2)(-3) + (-3)^2$

$4 +36 +9$

49

04 Given x=2, y=1

$5x+2y$

Substitute x=2, y=1.

$5\times 2 +2\times 1$

$10 +2$

12

Worked solutions

Multiple variables - mixed practice

05 Given a=-3, b=-4, c=1

$2a - 6b + 2c$

Substitute values.

$2\times (-3) - 6\times (-4) + 2\times 1$

$-6 + 24 + 2$

20

06 Given a=-3, b=1

$a^2 - 6ab + b^2$

Substitute a=-3, b=1.

$(-3)^2 - 6(-3)(1) + (1)^2$

$9 + 18 + 1$

28